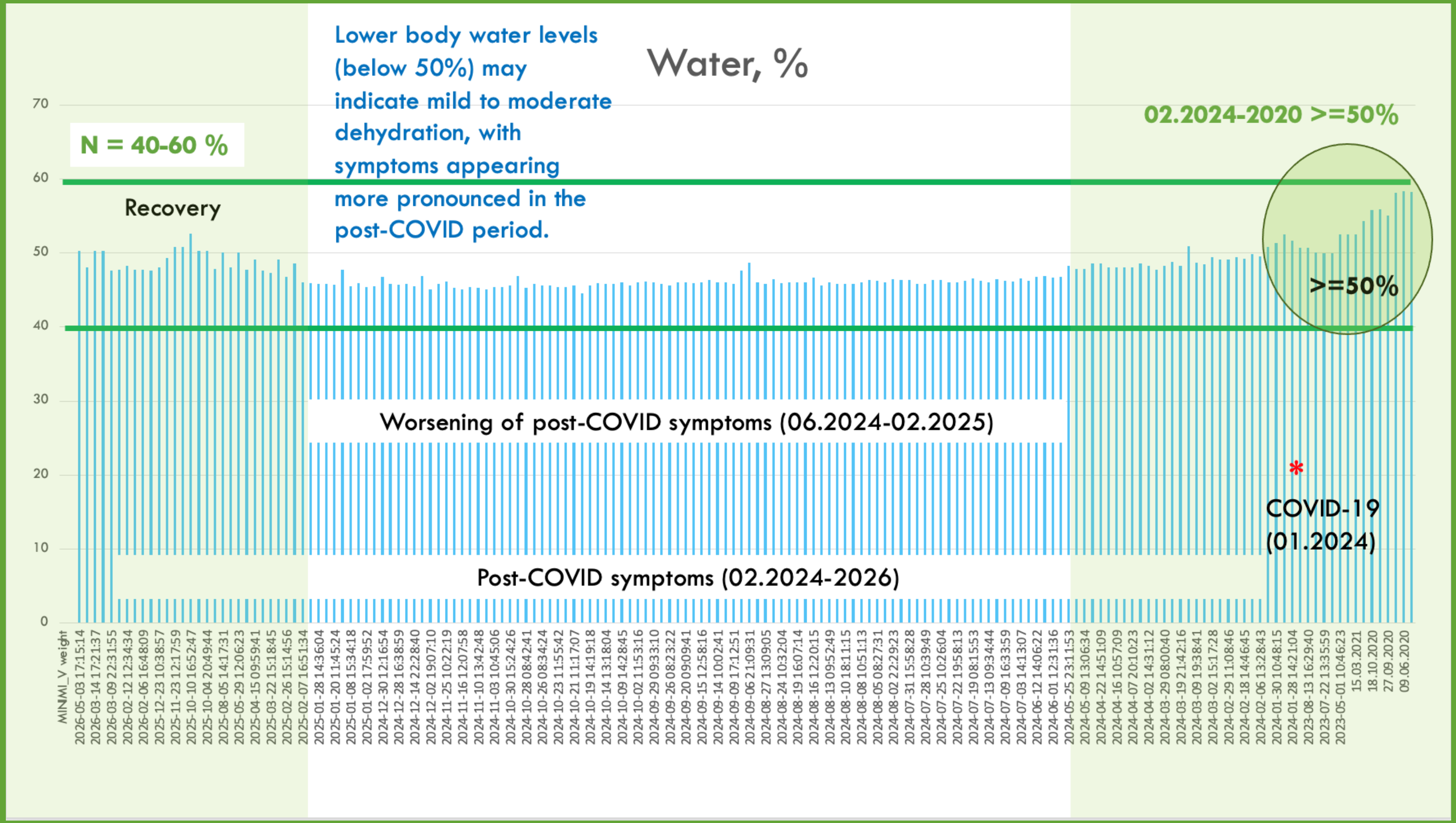
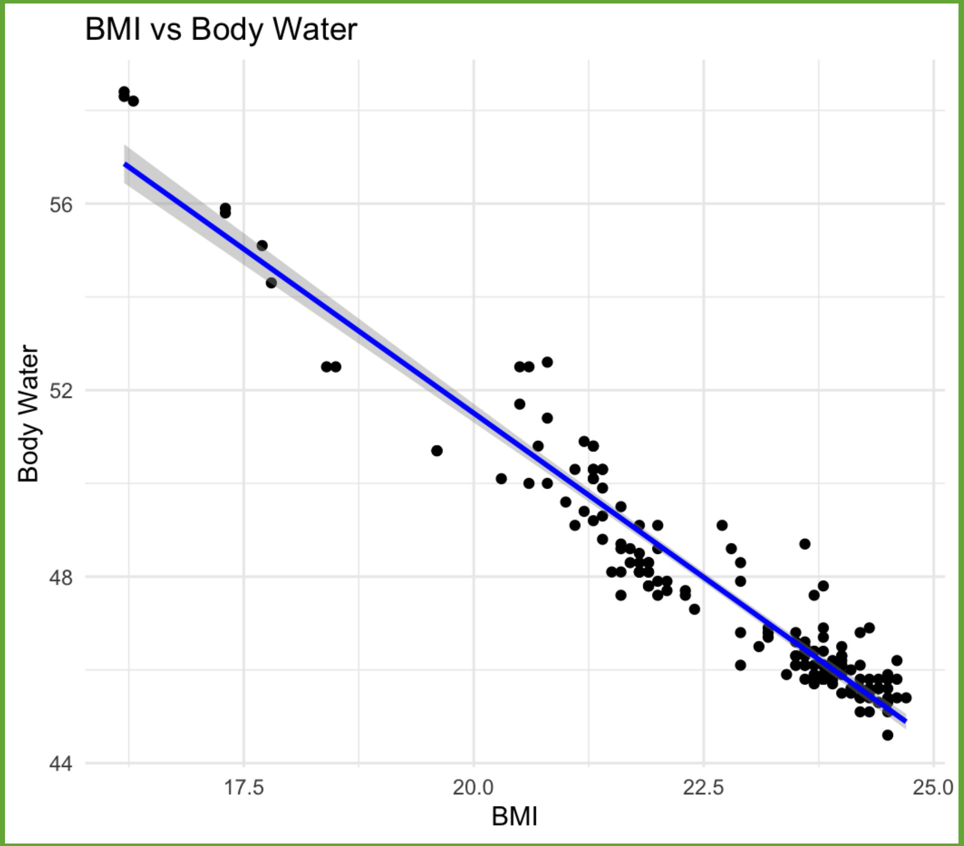




COVID-19 CLINICAL CASE WITH BODY COMPOSITION (BIA) ANALYSIS

Author: Valeriia Kainieva | 2026

COVID-19 CLINICAL CASE WITH BODY COMPOSITION (BIA) ANALYSIS

Exploratory analysis of longitudinal body composition (BIA) data during the COVID-19 period using R.

Analysis of repeated measurements revealed strong negative correlations between **body fat** and **body water** ($r = -0.99$, $p < 0.001$), as well as between **BMI** and **body water** ($r = -0.96$, $p < 0.001$), and a strong positive correlation between **BMI** and **body fat** ($r = 0.96$, $p < 0.001$).

As the data represent repeated measurements from **a single individual** over time, the observed relationships likely reflect underlying physiological dependencies and characteristics of BIA-derived measurements.

TOOLS AND TECHNOLOGIES

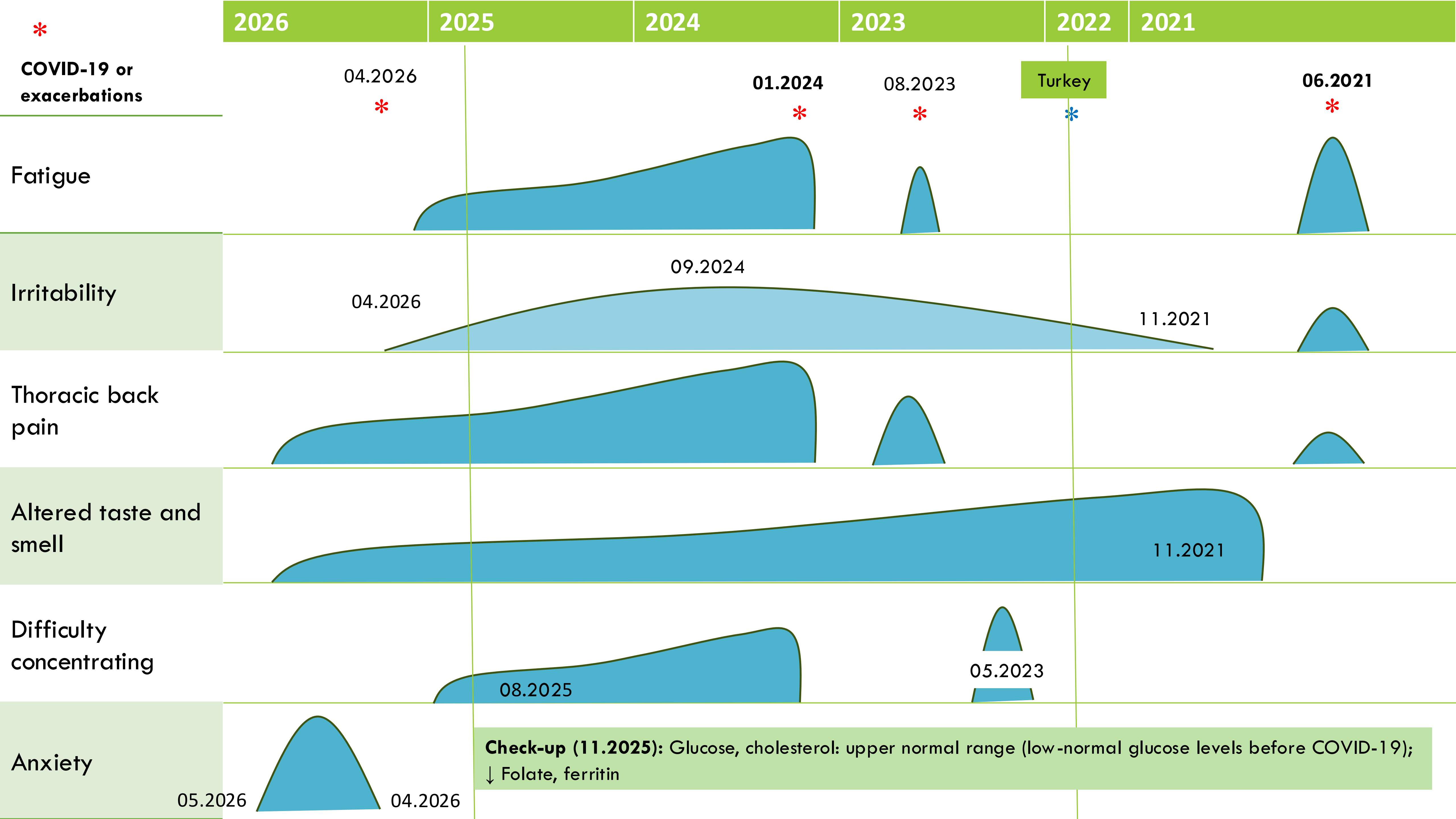
Noerden Minimi (bioelectrical impedance scale)

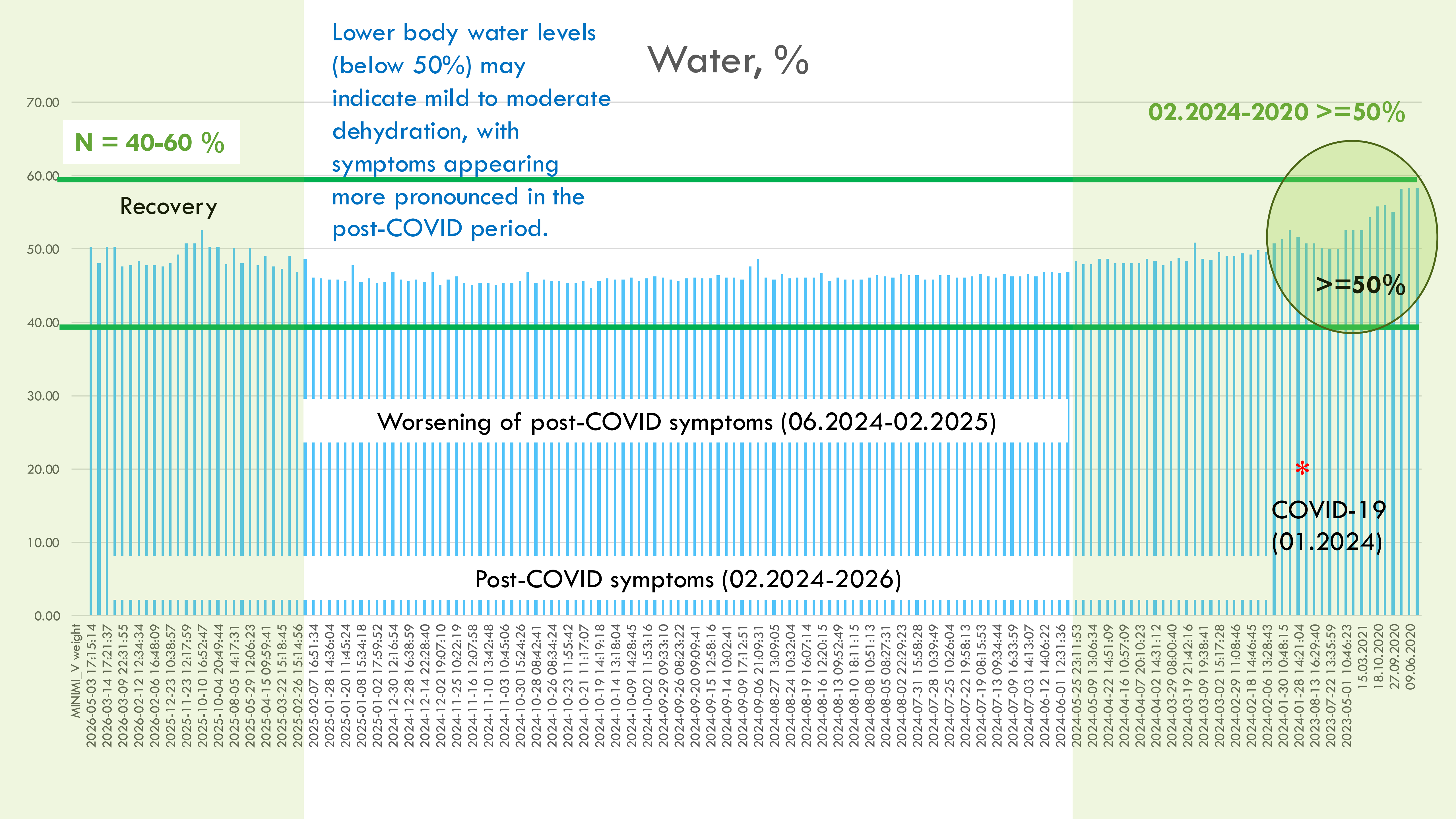
Microsoft Excel

Google Sheets

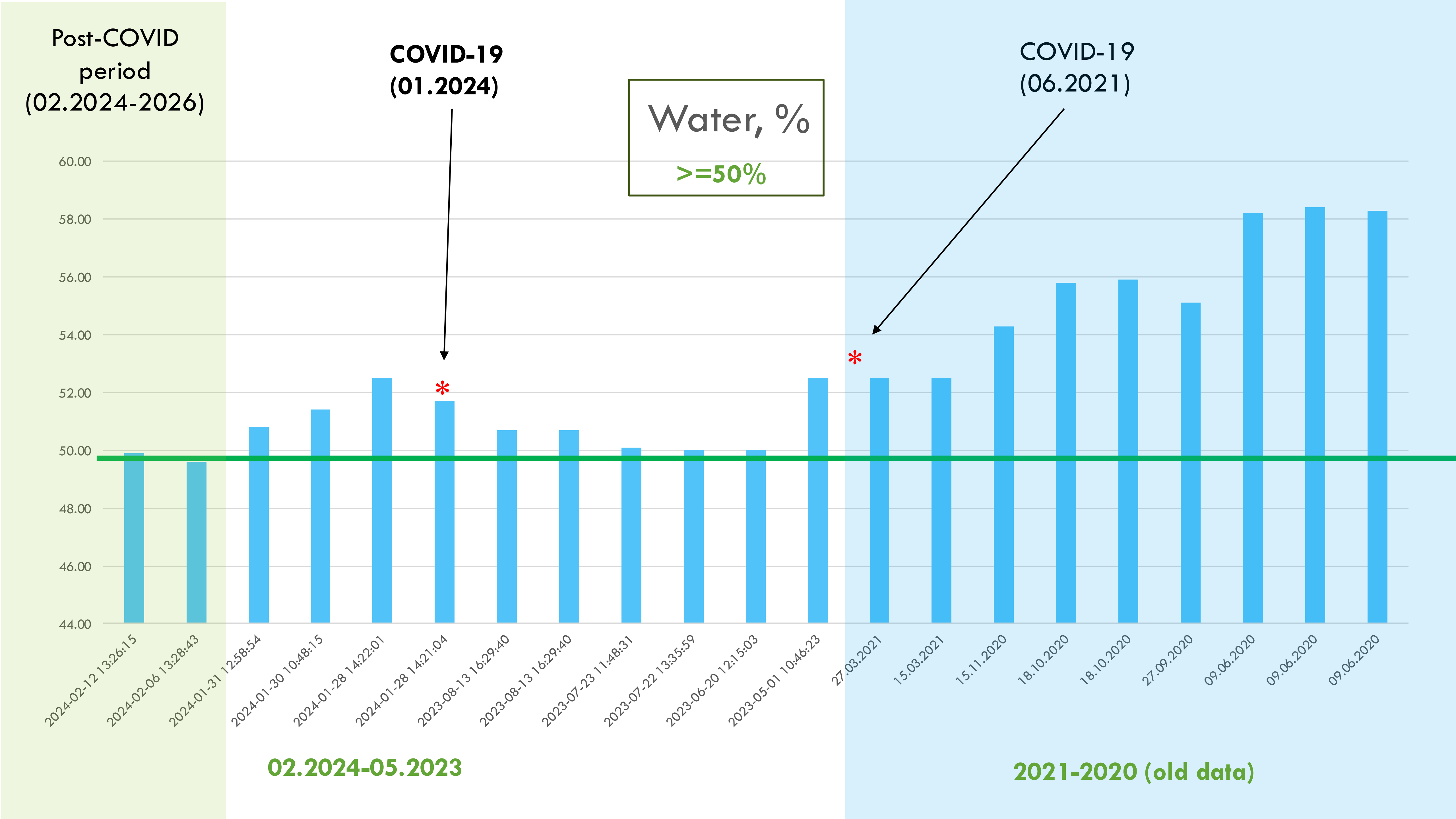
Microsoft PowerPoint

RStudio

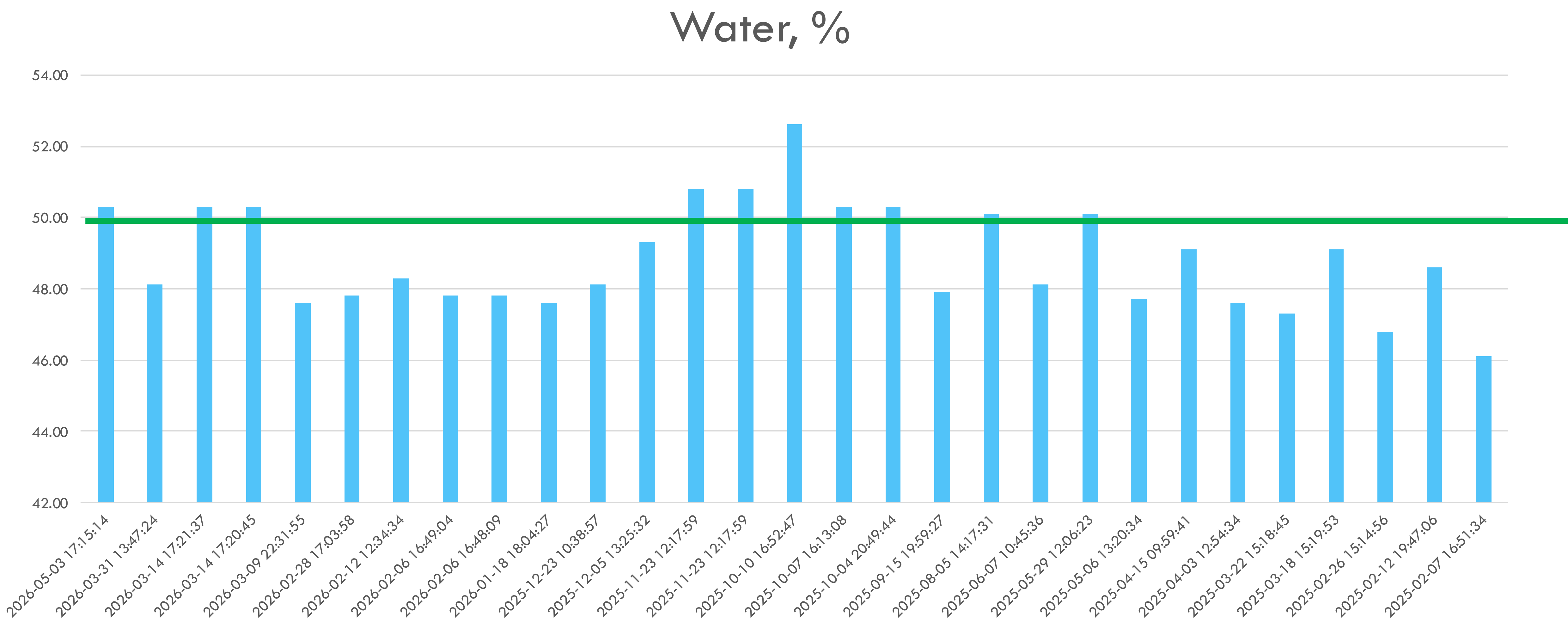

2026
2025
2024
2023
2022
2021
*
COVID-19 or exacerbations
04.2026
01.2024
08.2023
Turkey
06.2021
Fatigue
Irritability
09.2024
04.2026
11.2021
Thoracic back pain
Altered taste and smell
11.2021
Difficulty concentrating
05.2023
08.2025
Anxiety
05.2026
04.2026
Check-up (11.2025): Glucose, cholesterol: upper normal range (low-normal glucose levels before COVID-19);
↓ Folate, ferritin

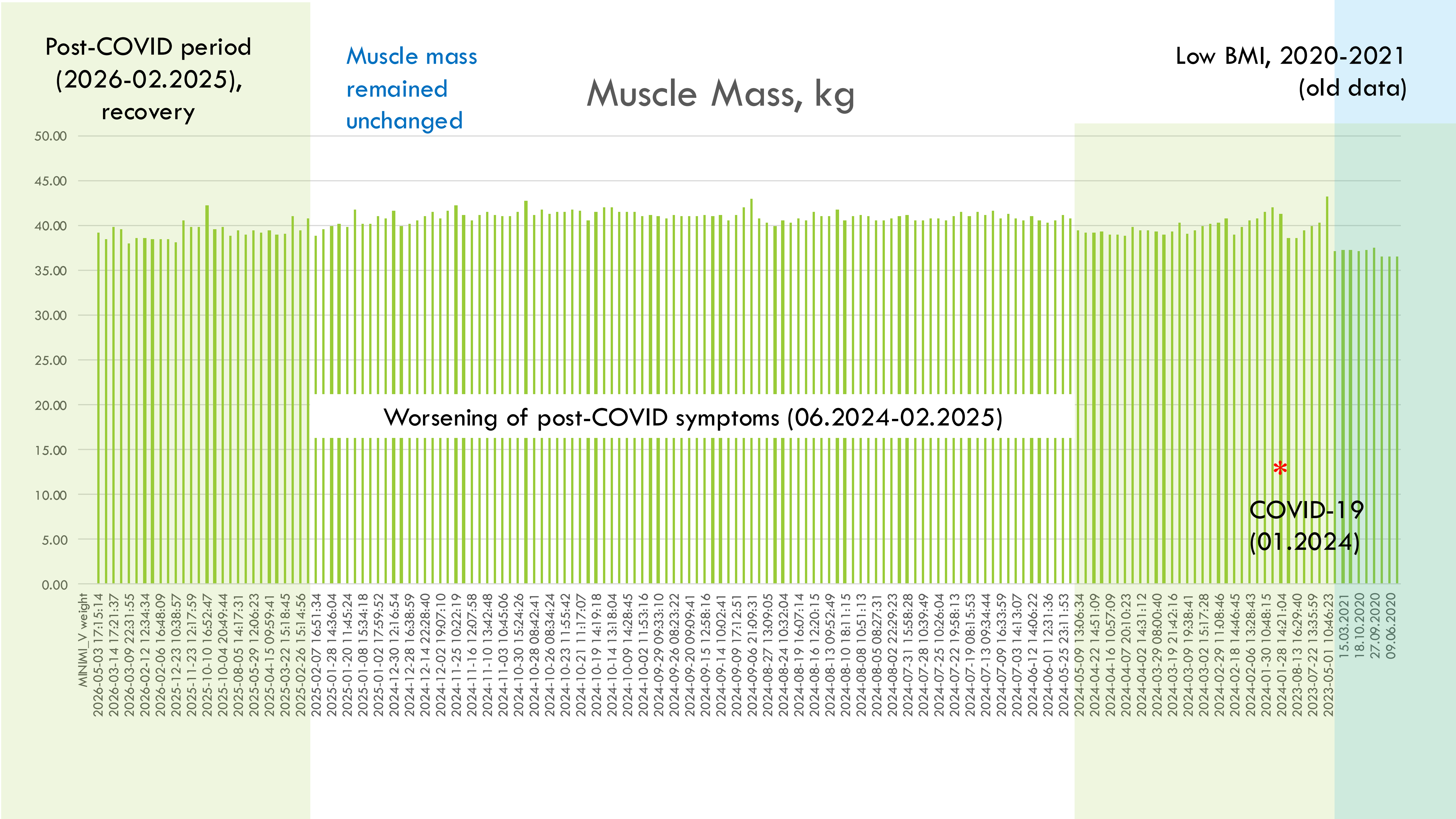


Water, %
Lower body water levels (below 50%) may indicate mild to moderate dehydration, with symptoms appearing more pronounced in the post-COVID period.
N = 40-60 %
02.2024-2020 >=50%
Recovery
>=50%
Worsening of post-COVID symptoms (06.2024-02.2025)
Post-COVID symptoms (02.2024-2026)
*
COVID-19 (01.2024)
70.00
60.00
50.00
40.00
30.00
20.00
10.00
0.00
MINIMI_V weight

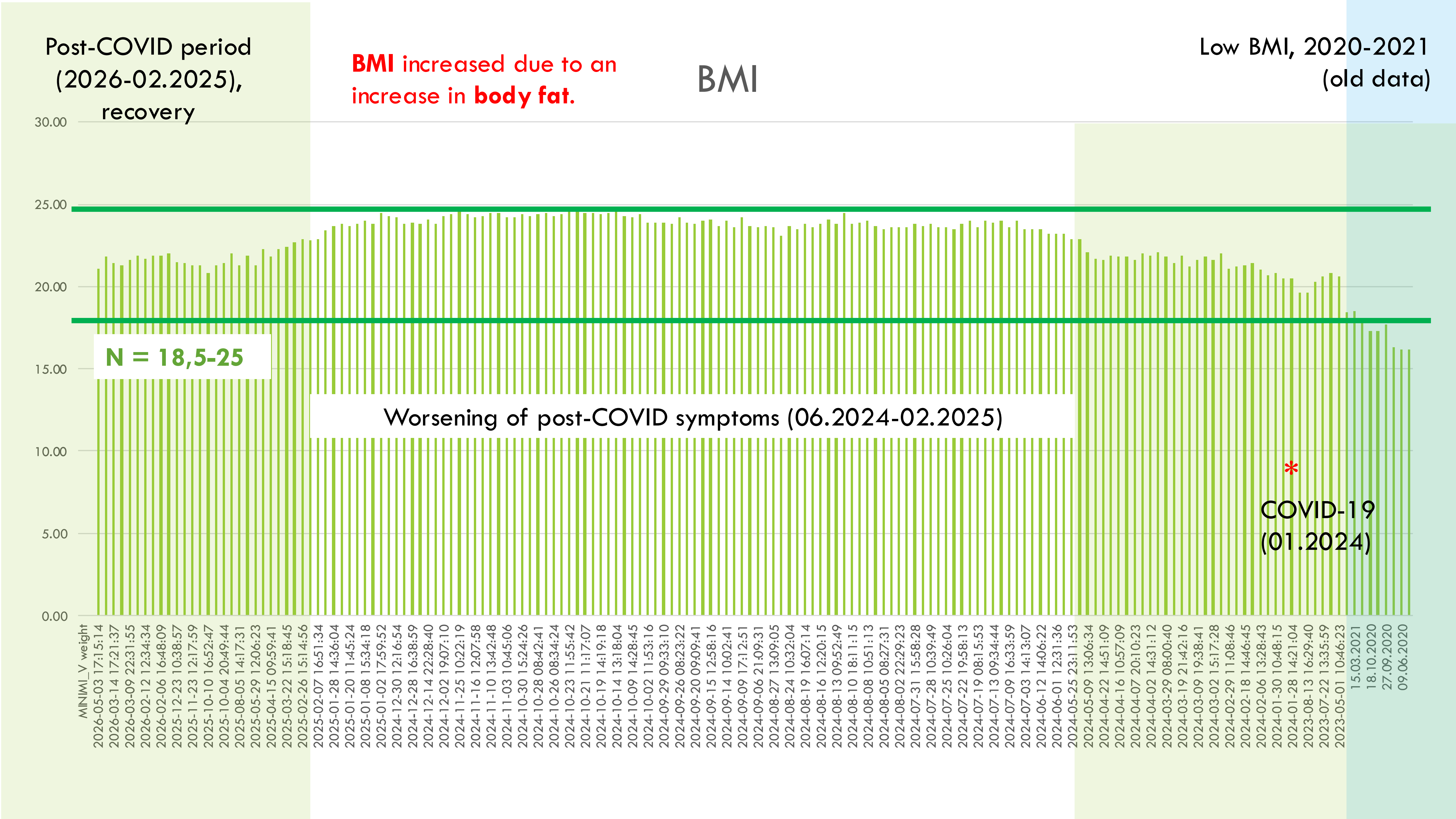

Post-COVID
period
(02.2024-2026)
COVID-19
(01.2024)
Water, %
>=50%
COVID-19
(06.2021)
60.00
58.00
56.00
54.00
52.00
50.00
48.00
46.00
44.00
2024-02-12 13:26:15
2024-02-06 13:28:43
2024-01-31 12:58:54
2024-01-30 10:48:15
2024-01-28 14:22:01
2024-01-28 14:21:04
2023-08-13 16:29:40
2023-08-13 16:29:40
2023-07-23 11:48:31
2023-07-22 13:35:59
2023-06-20 12:15:03
2023-05-01 10:46:23
27.03.2021
15.03.2021
15.11.2020
18.10.2020
18.10.2020
27.09.2020
09.06.2020
09.06.2020
09.06.2020
02.2024-05.2023
2021-2020 (old data)

Recovery after COVID-19 (2026-02.2025)





Post-COVID period (2026-02.2025), recovery
Muscle mass remained unchanged
Muscle Mass, kg
Low BMI, 2020-2021 (old data)
Worsening of post-COVID symptoms (06.2024-02.2025)
*
COVID-19 (01.2024)
50.00
45.00
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
MINIMI_V weight
2026-05-03 17:15:14
2026-03-14 17:21:37
2026-03-09 22:31:55
2026-02-12 12:34:34
2026-02-06 16:48:09
2025-12-23 10:38:57
2025-11-23 12:17:59
2025-10-10 16:52:47
2025-10-04 20:49:44
2025-08-05 14:17:31
2025-05-29 12:06:23
2025-04-15 09:59:41
2025-03-22 15:18:45
2025-02-26 15:14:56
2025-02-07 16:51:34
2025-01-28 14:36:04
2025-01-20 11:45:24
2025-01-08 15:34:18
2025-01-02 17:59:52
2024-12-30 12:16:54
2024-12-28 16:38:59
2024-12-14 22:28:40
2024-12-02 19:07:10
2024-11-25 10:22:19
2024-11-16 12:07:58
2024-11-10 13:42:48
2024-11-03 10:45:06
2024-10-30 15:24:26
2024-10-28 08:42:41
2024-10-26 08:34:24
2024-10-23 11:55:42
2024-10-21 11:17:07
2024-10-19 14:19:18
2024-10-14 13:18:04
2024-10-09 14:28:45
2024-10-02 11:53:16
2024-09-29 09:33:10
2024-09-26 08:23:22
2024-09-20 09:09:41
2024-09-15 12:58:16
2024-09-14 10:02:41
2024-09-09 17:12:51
2024-09-06 21:09:31
2024-08-27 13:09:05
2024-08-24 10:32:04
2024-08-19 16:07:14
2024-08-16 12:20:15
2024-08-13 09:52:49
2024-08-10 18:11:15
2024-08-08 10:51:13
2024-08-05 08:27:31
2024-08-02 22:29:23
2024-07-31 15:58:28
2024-07-28 10:39:49
2024-07-25 10:26:04
2024-07-22 19:58:13
2024-07-19 08:15:53
2024-07-13 09:34:44
2024-07-09 16:33:59
2024-07-03 14:13:07
2024-06-12 14:06:22
2024-06-01 12:31:36
2024-05-25 23:11:53
2024-05-09 13:06:34
2024-04-22 14:51:09
2024-04-16 10:57:09
2024-04-07 20:10:23
2024-04-02 14:31:12
2024-03-29 08:00:40
2024-03-19 21:42:16
2024-03-09 19:38:41
2024-03-02 15:17:28
2024-02-29 11:08:46
2024-02-18 14:46:45
2024-02-06 13:28:43
2024-01-30 10:48:15
2024-01-28 14:21:04
2023-08-13 16:29:40
2023-07-22 13:35:59
2023-05-01 10:46:23
15.03.2021
18.10.2020
27.09.2020
09.06.2020



Post-COVID period
(2026-02.2025),
recovery
BMI increased due to an
increase in body fat.
BMI
Low BMI, 2020-2021
(old data)
30.00
25.00
20.00
15.00
10.00
5.00
0.00
N = 18,5-25
Worsening of post-COVID symptoms (06.2024-02.2025)
*
COVID-19
(01.2024)
MINIMI_V weight
2026-05-03 17:15:14
2026-03-14 17:21:37
2026-03-09 22:31:55
2026-02-12 12:34:34
2026-02-06 16:48:09
2025-12-23 10:38:57
2025-11-23 12:17:59
2025-10-10 16:52:47
2025-10-04 20:49:44
2025-08-05 14:17:31
2025-05-29 12:06:23
2025-04-15 09:59:41
2025-03-22 15:18:45
2025-02-26 15:14:56
2025-02-07 16:51:34
2025-01-28 14:36:04
2025-01-20 11:45:24
2025-01-08 15:34:18
2025-01-02 17:59:52
2024-12-30 12:16:54
2024-12-28 16:38:59
2024-12-14 22:28:40
2024-12-02 19:07:10
2024-11-25 10:22:19
2024-11-16 12:07:58
2024-11-10 13:42:48
2024-11-03 10:45:06
2024-10-30 15:24:26
2024-10-28 08:42:41
2024-10-26 08:34:24
2024-10-23 11:55:42
2024-10-21 11:17:07
2024-10-19 14:19:18
2024-10-14 13:18:04
2024-10-09 14:28:45
2024-10-02 11:53:16
2024-09-29 09:33:10
2024-09-26 08:23:22
2024-09-20 09:09:41
2024-09-15 12:58:16
2024-09-14 10:02:41
2024-09-09 17:12:51
2024-09-06 21:09:31
2024-08-27 13:09:05
2024-08-24 10:32:04
2024-08-19 16:07:14
2024-08-16 12:20:15
2024-08-13 09:52:49
2024-08-10 18:11:15
2024-08-08 10:51:13
2024-08-05 08:27:31
2024-08-02 22:29:23
2024-07-31 15:58:28
2024-07-28 10:39:49
2024-07-25 10:26:04
2024-07-22 19:58:13
2024-07-19 08:15:53
2024-07-13 09:34:44
2024-07-09 16:33:59
2024-07-03 14:13:07
2024-06-12 14:06:22
2024-06-01 12:31:36
2024-05-25 23:11:53
2024-05-09 13:06:34
2024-04-22 14:51:09
2024-04-16 10:57:09
2024-04-07 20:10:23
2024-04-02 14:31:12
2024-03-29 08:00:40
2024-03-19 21:42:16
2024-03-09 19:38:41
2024-03-02 15:17:28
2024-02-29 11:08:46
2024-02-18 14:46:45
2024-02-06 13:28:43
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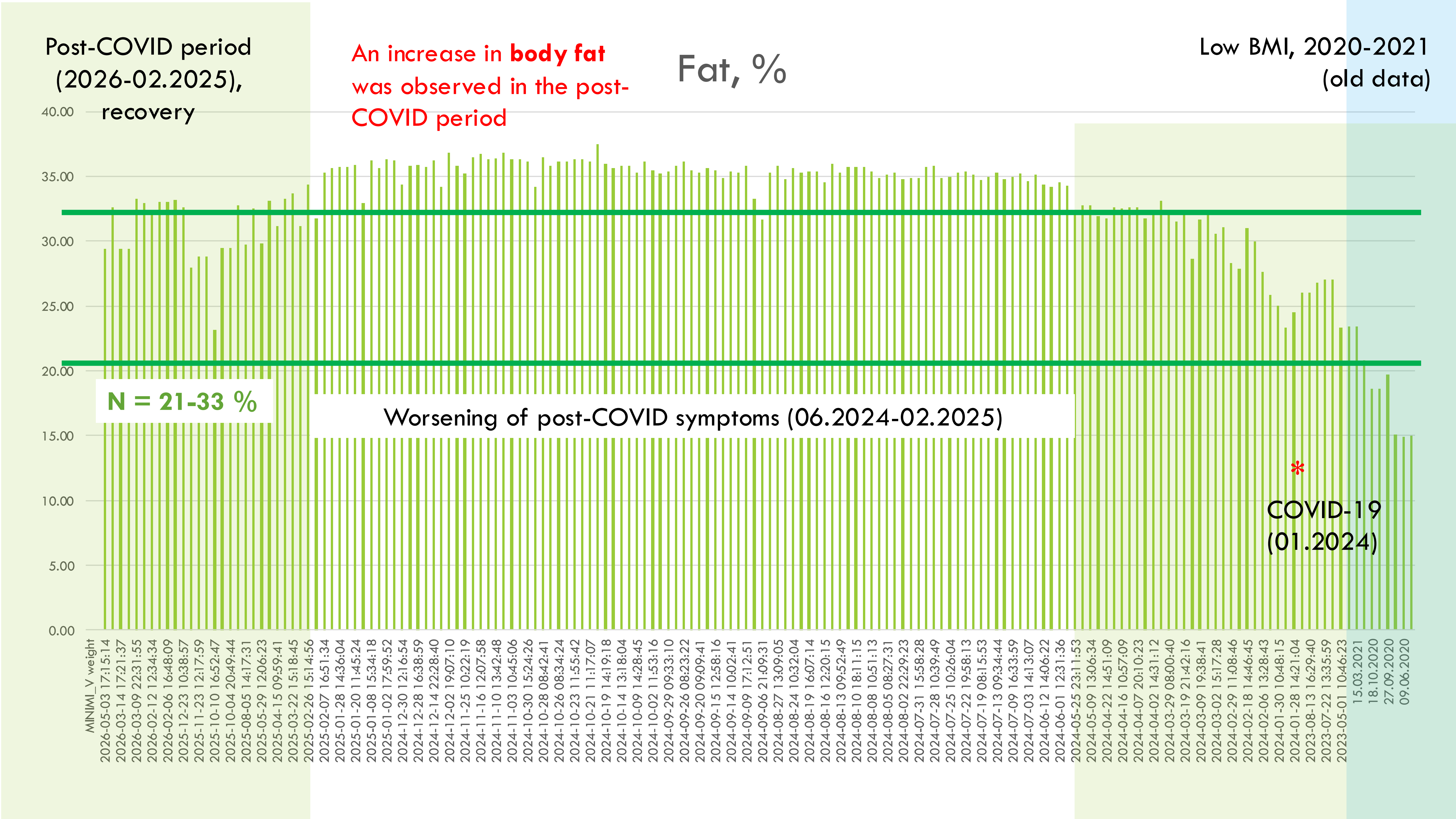
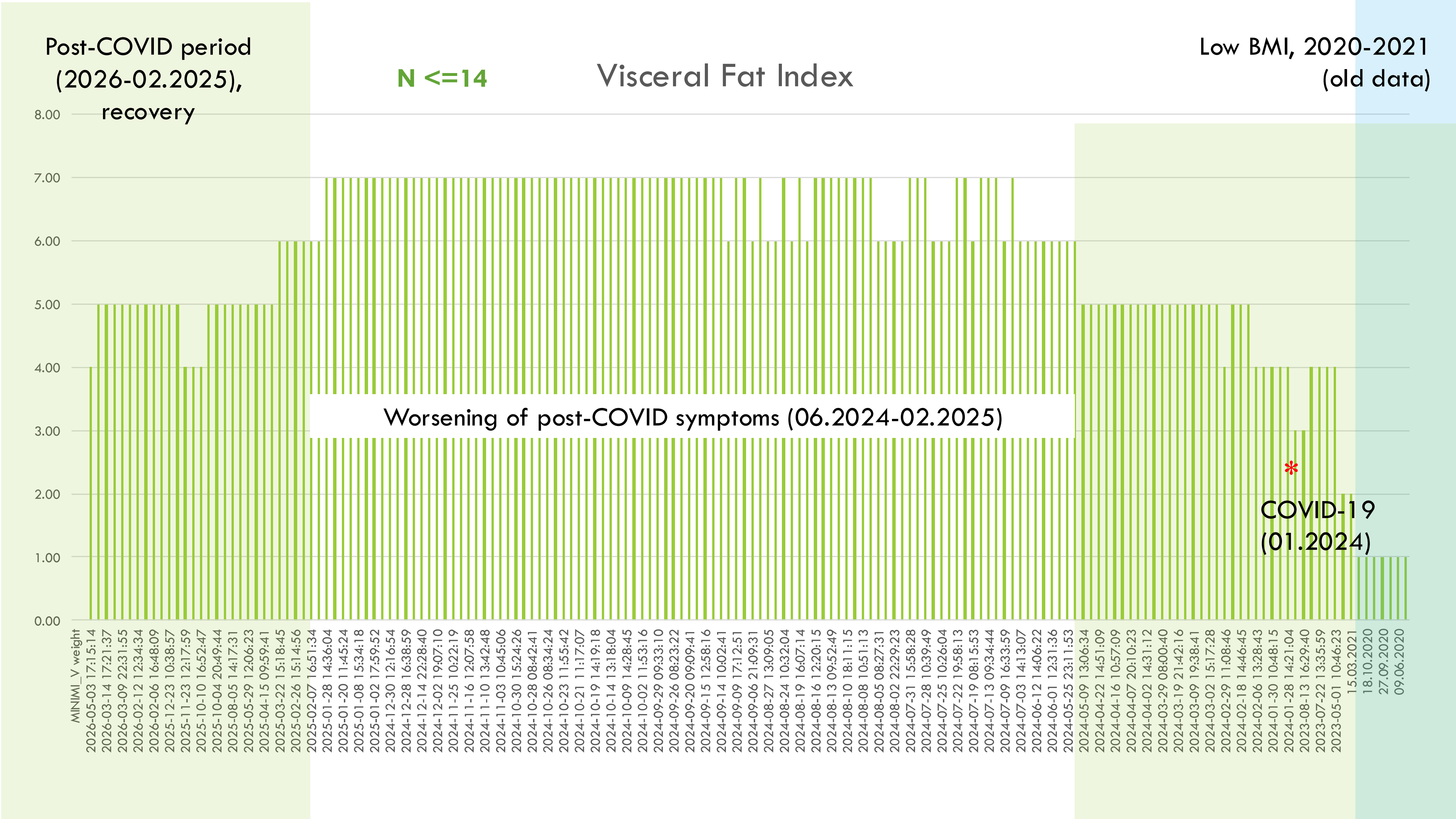
2024-01-28 14:21:04
2023-08-13 16:29:40
2023-07-22 13:35:59
2023-05-01 10:46:23
15.03.2021
18.10.2020
27.09.2020
09.06.2020


Post-COVID period (2026-02.2025), recovery
An increase in **body fat** was observed in the post-COVID period
Fat, %
Low BMI, 2020-2021 (old data)
N = 21-33 %
Worsening of post-COVID symptoms (06.2024-02.2025)
*
COVID-19 (01.2024)
40.00
35.00
30.00
25.00
20.00
15.00
10.00
5.00
0.00
MINIMI_V weight


Post-COVID period
(2026-02.2025),
recovery
N <=14
Visceral Fat Index
Low BMI, 2020-2021
(old data)
8.00
7.00
6.00
5.00
4.00
3.00
2.00
1.00
0.00
Worsening of post-COVID symptoms (06.2024-02.2025)
*
COVID-19
(01.2024)
MINIMI_V weight
2026-05-03 17:15:14
2026-03-14 17:21:37
2026-03-09 22:31:55
2026-02-12 12:34:34
2026-02-06 16:48:09
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2025-11-23 12:17:59
2025-10-10 16:52:47
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2024-11-16 12:07:58
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2024-10-28 08:42:41
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2024-10-23 11:55:42
2024-10-21 11:17:07
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2024-10-14 13:18:04
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2024-10-02 11:53:16
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2024-09-26 08:23:22
2024-09-20 09:09:41
2024-09-15 12:58:16
2024-09-14 10:02:41
2024-09-09 17:12:51
2024-09-06 21:09:31
2024-08-27 13:09:05
2024-08-24 10:32:04
2024-08-19 16:07:14
2024-08-16 12:20:15
2024-08-13 09:52:49
2024-08-10 18:11:15
2024-08-08 10:51:13
2024-08-05 08:27:31
2024-08-02 22:29:23
2024-07-31 15:58:28
2024-07-28 10:39:49
2024-07-25 10:26:04
2024-07-22 19:58:13
2024-07-19 08:15:53
2024-07-13 09:34:44
2024-07-09 16:33:59
2024-07-03 14:13:07
2024-06-12 14:06:22
2024-06-01 12:31:36
2024-05-25 23:11:53
2024-05-09 13:06:34
2024-04-22 14:51:09
2024-04-16 10:57:09
2024-04-07 20:10:23
2024-04-02 14:31:12
2024-03-29 08:00:40
2024-03-19 21:42:16
2024-03-09 19:38:41
2024-03-02 15:17:28
2024-02-29 11:08:46
2024-02-18 14:46:45
2024-02-06 13:28:43
2024-01-30 10:48:15
2024-01-28 14:21:04
2023-08-13 16:29:40
2023-07-22 13:35:59
2023-05-01 10:46:23
15.03.2021
18.10.2020
27.09.2020
09.06.2020

2021 VS COVID-19

Before 2021:

Muscle Mass

↑ Water

↓ Body Fat

After COVID-19:

Muscle Mass

↓ Water

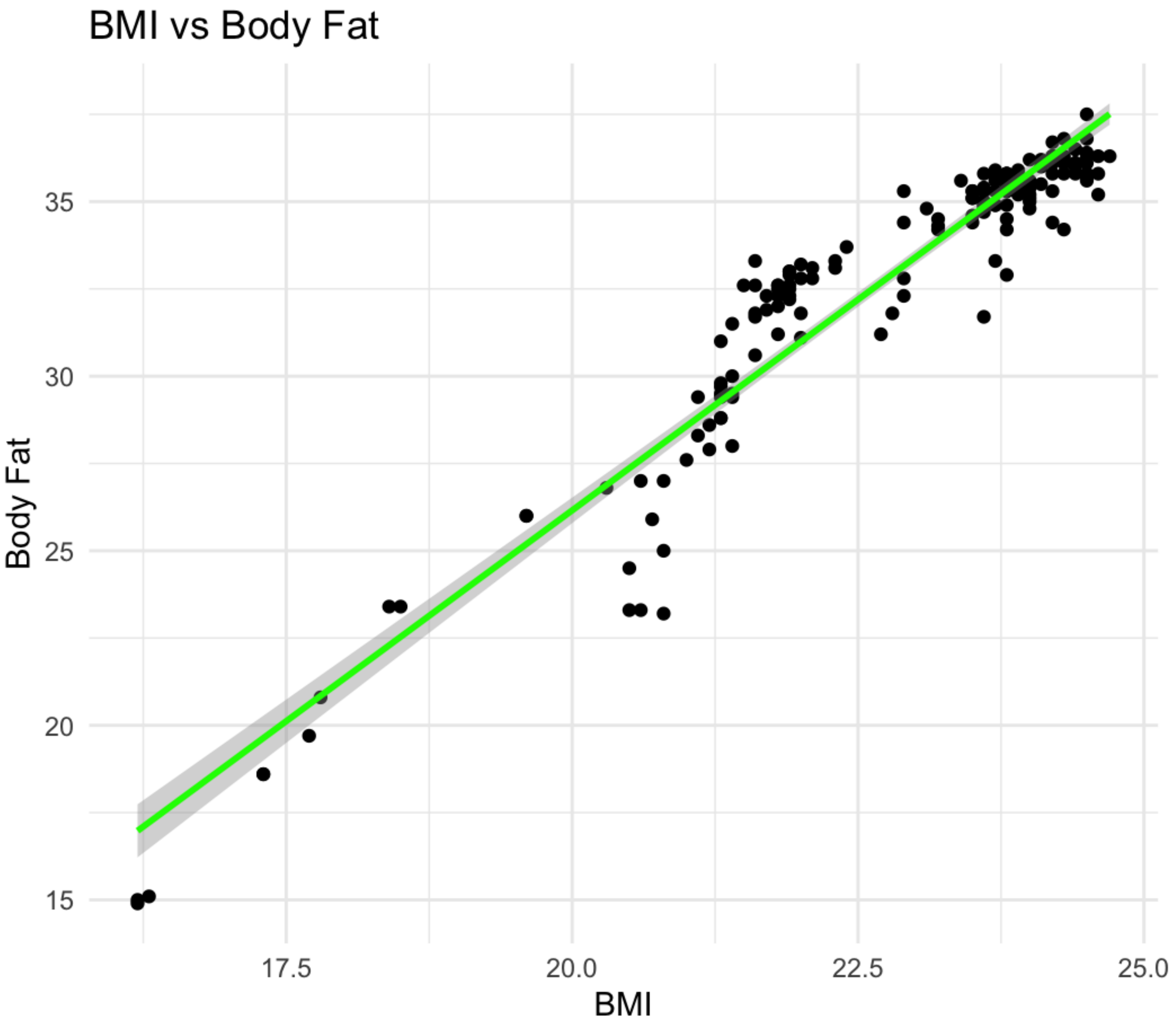
↑ Body Fat

"Muscle mass" values are device-derived estimates and should be interpreted as part of overall lean body mass.

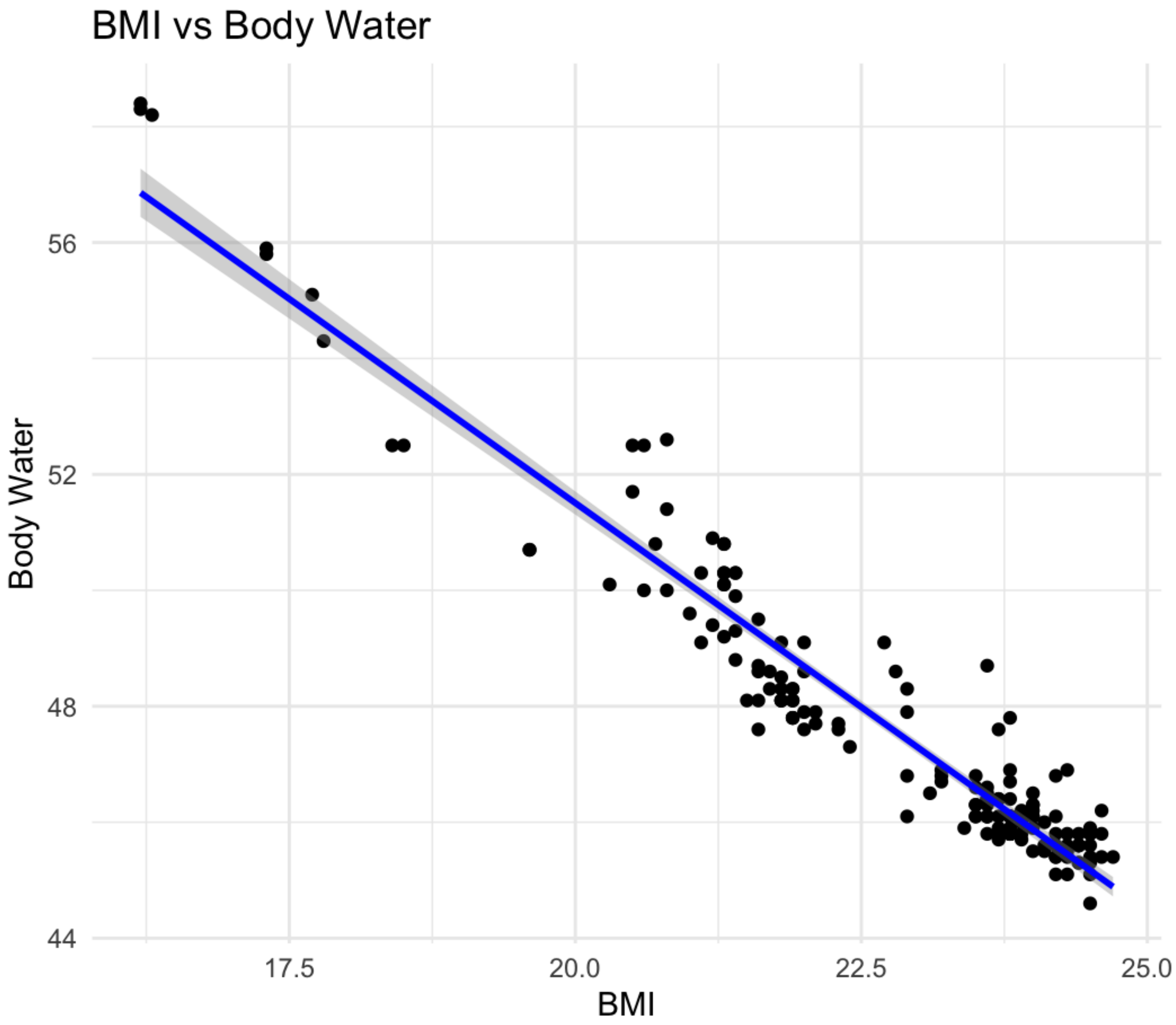
RSTUDIO

```
> data <- read.csv("data_bia.csv")
> names(data)
[1] "bmi"   "fat"   "water"
> str(data)
'data.frame':	168 obs. of  3 variables:
 $ bmi  : num  21.1 21.8 21.4 21.3 21.6 21.9 21.7 21.9 21.9 22 ...
 $ fat  : num  29.4 32.6 29.4 29.4 33.3 32.9 32.3 33 33 33.2 ...
 $ water: num  50.3 48.1 50.3 50.3 47.6 47.8 48.3 47.8 47.8 47.6 ...
> cor(data$bmi, data$fat)
[1] 0.9565598
> cor(data$fat, data$water)
[1] -0.9888601
> cor(data$bmi, data$water)
[1] -0.9613819
```

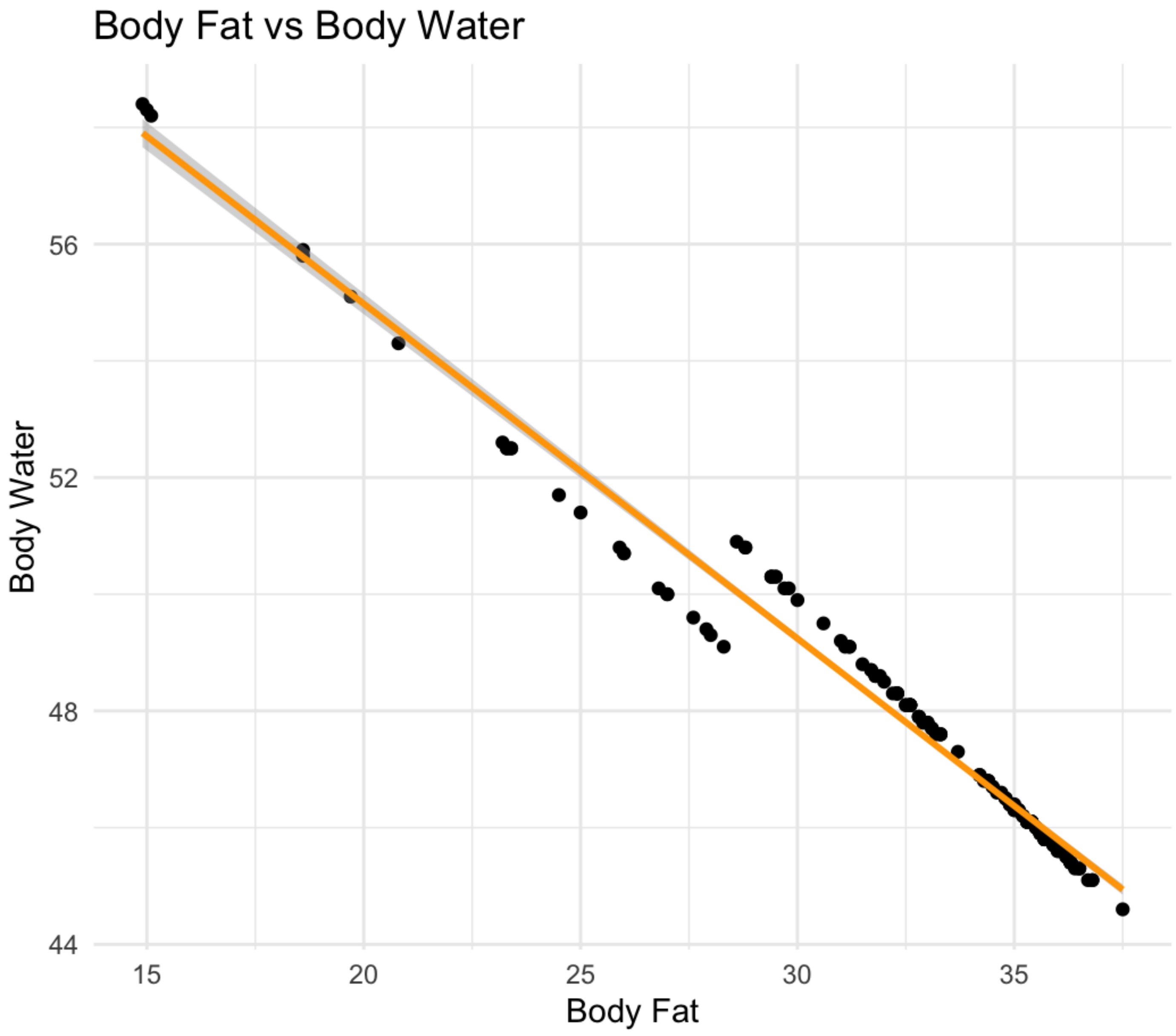
```
library(ggplot2)

ggplot(data, aes(x = bmi, y = fat)) +
  geom_point() +
  geom_smooth(method = "lm", color = ”green") +
  labs(title = "BMI vs Body Fat",
       x = "BMI",
       y = "Body Fat") +
  theme_minimal()
```



```
ggplot(data, aes(x = bmi, y = water)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  labs(title = "BMI vs Body Water",
       x = "BMI",
       y = "Body Water") +
  theme_minimal()
```



```
ggplot(data, aes(x = fat, y = water)) +
  geom_point() +
  geom_smooth(method = "lm", color = "blue") +
  labs(title = "Body Fat vs Body Water",
       x = "Body Fat",
       y = "Body Water") +
  theme_minimal()
```



```
cor.test(data$bmi, data$fat)

        Pearson's product-moment correlation

data:  data$bmi and data$fat
t = 42.274, df = 166, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9415115 0.9678006
sample estimates:
      cor
0.9565598
```

A very strong positive correlation was observed between **BMI** and **body fat** ($r = 0.96$, $p < 0.001$).

cor.test(data$bmi, data$water)

Pearson's product-moment correlation

data: data$bmi and data$water

t = -45.006, df = 166, **p-value < 2.2e-16**

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.9713931 -0.9479596

sample estimates:

cor

-0.9613819

A very strong negative correlation was observed between **BMI** and **body water** (r = -0.96, p < 0.001).

```
cor.test(data$fat, data$water)


        Pearson's product-moment correlation


data:  data$fat and data$water
t = -85.595, df = 166, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.9917779 -0.9849148
sample estimates:
      cor
-0.9888601
```

A very strong negative correlation was observed between **body fat** and **body water** ($r = -0.99$, $p < 0.001$).